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Research Paper

Stromal caveolin-1 levels predict early DCIS progression to invasive breast cancer

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Key words: caveolin-1, DCIS, recurrence, progression

Here, we determined the possible association of stromal caveolin-1 (Cav-1) levels with DCIS recurrence and/or progression to invasive breast cancer. An initial cohort of 78 DCIS patients with follow-up data was examined. As ER-positivity was associated with recurrence, we focused our analysis on this subset of 56 patients. In this group, we observed that DCIS progressed to invasive breast cancer in ~14% of the patient population (8/56), in accordance with an expected progression rate of 12–15%. Nearly ninety percent of DCIS patients (7/8) that underwent recurrence to invasive breast cancer had reduced or absent levels of stromal Cav-1. Remarkably, an absence of stromal Cav-1 (score = 0) was specifically associated with early disease progression to invasive breast cancer, with reduced time to recurrence and higher recurrence rate. All DCIS patients with an absence of stromal Cav-1 underwent some form of recurrence (5/5) and the majority (4/5) underwent progression to invasive breast cancer. This represents an overall cumulative incidence rate of 100% for recurrence and 80% for progression. An absence of stromal Cav-1 in DCIS lesions was also specifically associated with the presence of inflammatory cells. Conversely, ninety-seven percent of ER(+) DCIS patients (35/36) with high levels of stromal Cav-1 (score = 2) did not show any invasive recurrence over the duration of follow-up (4–208 mo), and 89% of such patients are estimated to remain free of invasive recurrence, even after 15 y. Thus, determination of stromal Cav-1 levels may be a useful new biomarker for guiding the treatment of ER(+) DCIS patients.

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Introduction

Since the introduction of screening mammography there has been a dramatic change in the presentation of breast cancer. During the 1970s, DCIS (ductal carcinoma in situ) comprised only 1% of all breast cancers. With the increased use of screening mammography, the incidence of DCIS has increased substantially and in 2005 comprised 20–30% of breast carcinoma diagnosed in United States.[1,2] Breast conserving therapy with or without radiotherapy is the accepted treatment for most cases of DCIS. However, the local recurrence rate with such therapy ranges from 10 to 40% with half of these patients developing invasive carcinoma.[3] Tamoxifen, used as an adjuvant systemic therapy for DCIS, does not increase overall survival and data on reduction of local recurrence rates are conflicting.[4,5]

It has also been shown that not all untreated DCIS cases will progress to invasive carcinoma. In a recent report of patients with low-grade DCIS who underwent breast biopsy, 39% developed invasive carcinoma in the same breast quadrant 3–42 years later.[6] At least 18% of these DCIS patients also developed distant metastasis, indicating that even morphologically low-grade DCIS may lead to a fatal disease outcome.[6] The current classification for DCIS based on nuclear grade, architectural differentiation and presence of necrosis, does not adequately predict the likelihood of recurrence after breast conserving therapy.[7] Therefore, there is a critical need to identify novel predictors of DCIS progression and potential targets for therapy.

Recently, we showed that an absence of stromal Cav-1 expression in human breast cancers is a powerful single independent predictor of early disease recurrence, metastasis, tamoxifen-resistance and poor clinical outcome.[8] More specifically, an absence of stromal Cav-1 immunostaining was associated with early disease recurrence, advanced tumor stage and lymph node metastasis, resulting in an ~3.6-fold reduction in progression-free survival.[8] In lymph-node positive patients, an absence of stromal Cav-1 predicted an ~11.5-fold reduction in 5-year progression-free survival.[8] As such, we proposed that Cav-1 may function as a tumor suppressor in the stromal micro-environment.[9] However, it remains unknown

whether stromal Cav-1 levels play any role in the invasive progression of "pre-malignant" lesions, such as ductal carcinoma in situ (DCIS) of the breast.

The aim of this study was to evaluate the stromal expression of Cav-1 in a cohort of DCIS patients treated with wide-excision and close follow-up, but without any form of chemo- or radio-therapy. This would allow us to establish the predictive value of stromal Cav-1 during the natural history of DCIS disease progression. For this purpose, we specifically examined the association between stromal Cav-1 expression, clinico-pathological variables, and DCIS recurrence or progression to invasive breast cancer.

Our results indicate that an absence of stromal Cav-1 is a novel biomarker that predicts early DCIS progression to invasive breast cancer. Conversely, high levels of stromal Cav-1 appear to be protective against DCIS progression to invasive disease.

Results

Description of the DCIS Patient Cohort. An initial cohort of 78 DCIS patients with follow-up data was examined, and subjected to Cav-1 immunostaining (Table 1). Data on ER/PR/HER2 status, DCIS lesion type, nuclear grade, necrosis and the presence of inflammatory cells were available. Similarly, follow-up data on DCIS recurrence and DCIS progression to invasive breast cancer were also available. Patient DCIS lesions were then scored, depending on the expression levels of stromal Cav-1 (high, score = 2; low, score = 1; absent, score = 0). [See Materials and Methods for further details]. Representative examples are shown in Figure 1. A subset of DCIS lesions also showed the presence of inflammatory cells (Fig. 1E).

Based on our initial analysis, we noticed that none of the ER-negative (-) DCIS patients underwent recurrence. Thus, we focused our analysis on the remaining subset of 56 ER-positive (+) DCIS patients. In this patient population, the incidence of high, low and absent stromal Cav-1 immunostaining is 64% (36/56), 27% (15/56) and 9% (5/56), respectively. We note that overall, 89% (7/8) of DCIS recurrences occurred within 5 years of surgery, whereas 63% (5/8) of invasive recurrences occurred more than 5 years after surgery.

Association of stromal Cav-1 with overall recurrence. Of the 56 ER(+) DCIS patients, sixteen of them underwent some form of recurrence (8 recurred to DCIS, while the other 8 recurred with progression to invasive breast cancer), yielding an overall recurrence rate of twenty-eight percent (Table 2A). Thus, the rate of DCIS progression to invasive breast cancer in this cohort is fourteen percent (8/56), consistent with the expected rate of 12–15%. The estimated 15 years recurrence rate overall was 35%, with a rate of 20.5% for invasive breast cancer recurrence and 14.9% for DCIS recurrence, based on Kaplan-Meier estimation adjusting for competing risks (Table 2B).

Interestingly, nearly seventy percent of these DCIS patients with any type of recurrence (11/16) had reduced levels of stromal Cav-1 (score = 1 or 0) (Tables 2A and 3).

No prognostic associations were observed with patient age at diagnosis, nuclear grade, necrosis, PR status, or HER2 status (See Suppl. Tables S1–S3). However, a trend towards significance was noted for many of these variables, and especially for HER2 status.

Table 1 **Description of recurrence and time to recurrence (TTR, months) for patients with primary DCIS**

Recurrence	N	Median TTR	Range TTR
None	58		
DCIS	9	30.33	17.6–129.7
Invasive	11	102.9	4.3–199.3

Table 1 gives a description of the overall pattern of recurrence among all 78 patients. Sixteen out of the 20 recurrent patients have ER(+) primary DCIS, and the remaining four do not have information about their ER status. This makes a strong case for ER(+) status being predictive of future recurrence. We focused our analysis on the patients with ER(+) primary DCIS (56 patients), since we did not see any recurrent disease in patients with ER(-) primary DCIS. TTR, time to recurrence (mo).

Association of stromal Cav-1 with DCIS progression to invasive breast cancer. Fifty percent of the DCIS patients (4/8) that showed recurrence to DCIS had reduced levels of stromal Cav-1 (score = 1 or 0). So, reduced stromal Cav-1 levels are not predictive of recurrence to DCIS per se, but may still be predictive of recurrence to invasive breast cancer (Table 4). In this regard, nearly ninety percent of DCIS patients (7/8) that underwent recurrence to invasive breast cancer had reduced levels of stromal Cav-1 (score = 1 or 0) (Tables 2A and 5).

Ninety-seven percent of DCIS patients (35/36) with high levels of stromal Cav-1 (score = 2) did not show recurrence to invasive breast cancer; only one of these patients showed invasive progression (Table 2A). In striking contrast, eighty percent of DCIS patients (4/5) with an absence of stromal Cav-1 (score = 0) showed recurrence to invasive breast cancer. In these patients (score = 0), median time to invasive recurrence (TTR) was also significantly reduced by nearly two-fold (170.57 versus 89.3 months) (Table 2A). The 15 year recurrence rate for invasive breast cancer was 11% for patients with high stromal Cav-1 (score = 2), compared with 60% for patients with an absence of Cav-1 (score = 0) (Table 2B). This means that 89% of patients with high stromal Cav-1 levels are expected to be recurrence-free after even 15 years, compared to only 40% of patients with no stromal Cav-1.

Kaplan-Meier curves for overall recurrence, recurrence to DCIS, and progression to invasive breast cancer are shown in Figures 2–4, respectively. Note that an absence of stromal Cav-1 is specifically associated with DCIS progression to invasive breast cancer, rather than simply DCIS recurrence (compare Tables 3–5).

Comparative effects of stromal Cav-1 on cumulative incidence of recurrence and progression rates. An absence of stromal Cav-1 increased the overall cumulative incidence of recurrence (DCIS + Invasive) by ~7-fold (14% (5/36) for score = 2 versus 100% (5/5) for score = 0) (Table 2A).

Similarly, a reduction in stromal Cav-1 increased the cumulative incidence of progression to invasive breast cancer by ~12-fold (3% (1/36) for score = 2 versus 35% (7/20) for score = 1 or 0). Finally, an absence of stromal Cav-1 increased the cumulative incidence of

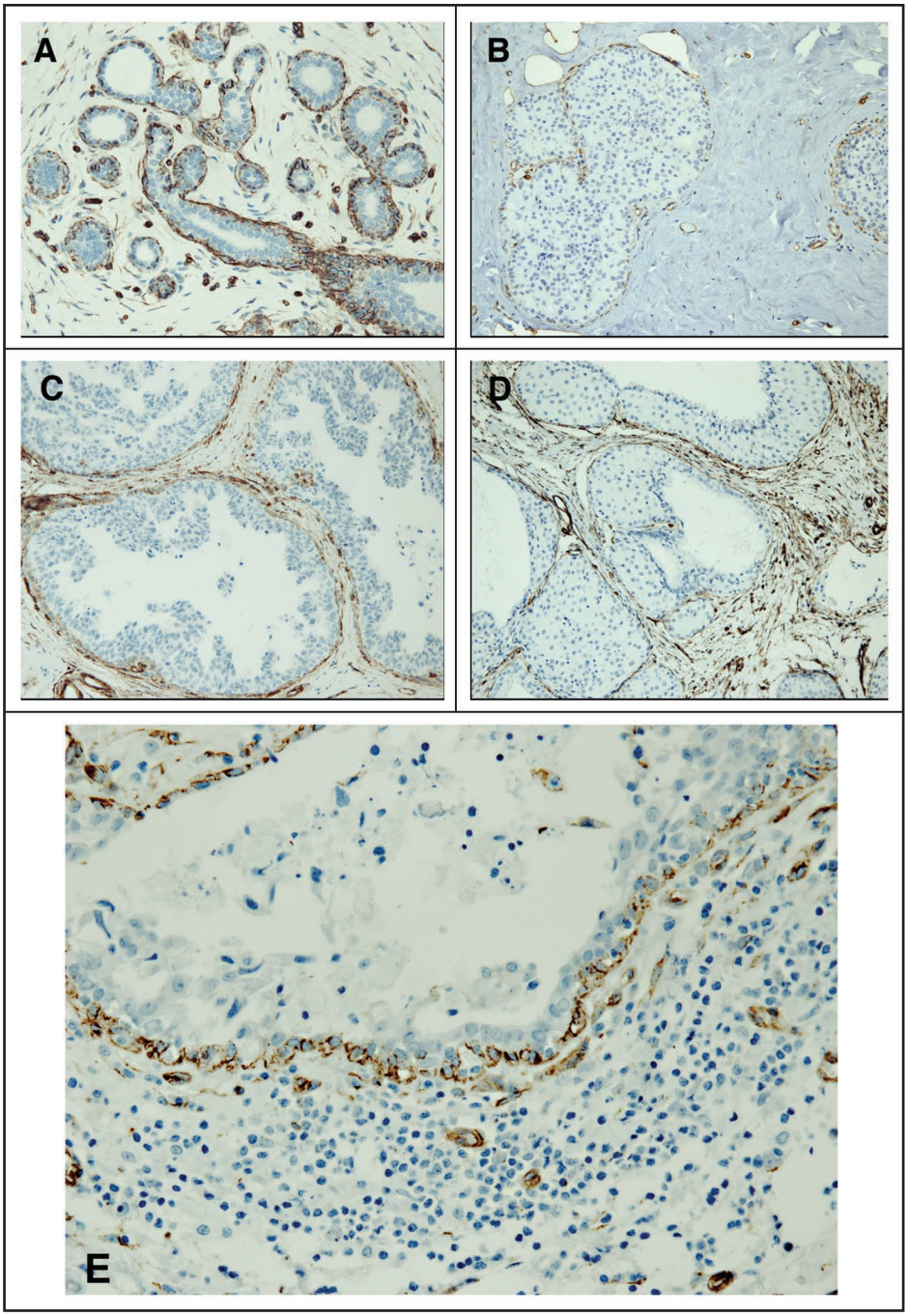

Figure 1. Cav-1 Immunostaining in Normal Breast and Ductal Carcinoma In Situ (DCIS). Representative examples are shown. (A) Focal weak staining in stromal fiboblasts, in normal terminal duct lobular units. (B) Score = 0 (no stromal Cav-1 staining) in DCIS. (C) Weak expression of stromal Cav-1 (Score = 1) in DCIS. (D) Strong diffuse staining in stromal fibroblasts in DCIS, representing Score = 2. (E) Loss of stromal Cav-1 expression in DCIS, in an area of inflammation. In all panels, note that myoepithelial and endothelial cells exhibit strong staining (original magnification: 40X objective (A and E) and 20X objective (B–D)).

progression to invasive breast cancer by ~27-fold (3% (1/36) for score = 2 versus 80% (4/5) for score = 0) (Table 2A).

Thus, patients with an absence of stromal Cav-1 (score = 0) represent only 9% (5/56) of the ER(+) DCIS population, but represent 50% (4/8) of the patients with invasive progression, and have a progression rate of 80% (4/5). As such, an absence of stromal Cav-1 is a very strong and selective predictor of DCIS progression to invasive breast cancer (Fig. 4 and Table 5).

Cox proportional hazards regression of incidence of invasive recurrence related to stromal Cav-1 expression levels. Table 6 shows the effect of stromal Cav-1 expression on the risk of developing invasive breast cancer after a diagnosis of ER(+) DCIS. Loss of stromal Cav-1 is strongly associated with an increased risk of recurrence to invasive breast cancer, using a Cox multivariate analysis which includes TTR. The estimated hazard ratio for patients with complete loss of stromal Cav-1 (score = 0) predicts a ~34-fold increased risk of developing progression to invasive breast cancer. However, since this is based on a relatively small number of cases, the 95% confidence interval is wide. However, there is significantly increased risk in this group, since the lower bound of the 95% confidence interval for the hazard ratio is 3.73. In fact, the lower bound of the 99% confidence interval is 2.46. This indicates that this group of patients has a minimum of a 2.5-fold increased risk for invasive recurrence to breast cancer.

Similar results were obtained when patients were grouped as high (score = 2) and low (score = 0 or 1) stromal Cav-1 (Table 6), resulting in a hazard ratio of ~14, with a much narrower 95% confidence interval.

DCIS patients with absent stromal Cav-1 all show infiltration with inflammatory cells. We also noted that inflammation (the presence of lymphocytes and/or macrophages) was specifically associated with DCIS progression to invasive breast cancer (Table 4 and Fig. 5). However, this effect was strictly dependent on stromal Cav-1 levels (Tables 7 and 8; Fig. 6).

For example, DCIS patients with high levels of stromal Cav-1 (score = 2) and inflammation did not show any recurrence (N = 6) (Table 7 and Fig. 6). In contrast, DCIS patients with an absence of stromal Cav-1 (score = 0), all underwent recurrence (1/5) or progression (4/5) and all showed inflammatory infiltration (5/5).

Thus, there is an emerging relationship between loss of stromal Cav-1, inflammation and DCIS recurrence/progression.

Discussion

Here, we show that ninety-seven percent of ER(+) DCIS patients (35/36) with high levels of stromal Cav-1 (score = 2) did not show any recurrence to invasive breast cancer. Moreover, 89% of patients with high levels of stromal Cav-1 are expected to be free of recurrent invasive carcinoma, even after 15 years. Almost ninety percent of DCIS patients (7/8) that underwent recurrence to invasive breast cancer had reduced levels of stromal Cav-1 (score = 1 or 0). Finally, eighty percent of DCIS patients (4/5) with an absence of stromal Cav-1 (score = 0) showed recurrence to invasive breast cancer; their median time to invasive recurrence (TTR) was also significantly reduced by nearly two-fold (170.57 versus 89.3 mo).

Thus, determination of stromal Cav-1 levels may be a useful new biomarker for guiding the treatment of ER(+) DCIS patients. Since high stromal Cav-1 levels are a strong predictor of favorable clinical outcome (an absence of progression to invasive breast cancer), its wide-spread use could spare nearly two-thirds (36/56) of the ER(+) DCIS population from over-treatment. This would allow clinicians to focus on the treatment of more high-risk patients with reduced or absent stromal Cav-1 (20/56), which represent nearly ninety-percent (7/8) of the DCIS patients that underwent progression to invasive breast cancer.

The associations we observe here between an absence of stromal Cav-1 and local inflammation may have important functional and biological significance. We have previously demonstrated that Cav-1 (-/-) mammary stromal fibroblasts behave like cancer-associated fibroblasts[13] and secrete or express elevated levels of a number of pro-inflammatory growth factors/cytokines (such as PDGF, VEGF, HGF, MIP2, MIP1-alpha, GM-CSF, TARC, L-Selectin and IL-6/11/10/13)[13] involved in regulating lymphocyte, neutrophil and macrophage recruitment. Thus, an absence of stromal Cav-1 in mammary fibroblasts may exacerbate local inflammation, thereby promoting an EMT-like genetic program in neighboring epithelial cells via paracrine secreted factors, leading to DCIS recurrence and progression to invasive breast cancer. Consistent with this hypothesis, human breast cancer-associated fibroblasts show a loss of Cav-1 protein expression, as compared with matched normal human fibroblasts obtained from the same patients, in eight out of 11 breast cancer patients examined.[14]

Interestingly, Cav-1 (-/-) mice develop high-grade DCIS-like lesions, with micro-invasion (as evidenced by a discontinuous myo-epithelial cell layer), and extensive mammary stromal angiogenesis, when challenged with estrogen.[15] As Cav-1 is not normally expressed in luminal mammary epithelial cells, while mammary stromal fibroblasts abundantly express Cav-1, this DCIS-like phenotype observed in estrogen-stimulated Cav-1 (-/-) null mice may be a "stromal phenomenon". Consistent with this hypothesis, we have previously shown using mammary transplant analysis that the ductal thickening (a.k.a., hyperplasia) phenotype we observed in Cav-1 (-/-) null mice is strictly dependent on the Cav-1 (-/-) mammary stroma.[16] More specifically, when Cav-1 (+/+) mammary epithelial cells were implanted into cleared Cav-1 (-/-) mammary stromal fat pads, an ~2-fold increase in ductal thickening was observed.[16] Similarly, orthotopic transplantation of Cav-1 (+/+) MMTV-PyMT tumor tissue into the mammary

Table 2 (A) Cumulative incidence of DCIS and breast cancer recurrence among ER(+) DCIS patients

	Stromal Cav-1 status		
	Score = 2 N = 36	Score = 1 N = 15	Score = 0 N = 5
Recurrence type			
None	86% (31)	60% (9)	0% (0)
DCIS	11% (4)	20% (3)	20% (1)
Invasive (Inv)*	3% (1)	20% (3)	80% (4)
Overall (DCIS + Inv)**	14% (5)	40% (6)	100% (5)
Median TTR (DCIS)	29.88	50.43	19.93
Median TTR (Invasive)***	170.57	102.9	89.3

*There is significant association between stromal Cav-1 expression and type of incident recurrence ($p = 1.3 \times 10^{-4}$, Fisher's exact test). **There is a significant association between stromal Cav-1 expression and incidence of disease recurrence (DCIS or invasive) ($p = 1.48 \times 10^{-4}$, Fisher's exact test). ***Time to recurrence (TTR, mo) also significantly differs for invasive recurrence by stromal Cav-1 expression (DCIS: $p = 0.509$; Invasive: $p = 3.1 \times 10^{-5}$, log-rank test).

Table 2 (B) 15 year (180 month) recurrence rates based on accounting for competing risks stratified by stromal Cav-1 expression within ER(+) DCIS patients

	Overall	Stromal Cav-1 status		
	N = 56	Score = 2 N = 36	Score = 1 N = 15	Score = 0 N = 5
Recurrence type				
DCIS	14.85%	12%	20%	20%
Invasive breast cancer (Inv)*	20.55%	11%	27.9%	60%
Overall (DCIS + Inv)**	35.4%	23%	47.9%	80%

*There is a significant association between TTR of invasive breast cancer and stromal Cav-1 expression ($p = 3.1 \times 10^{-5}$; log-rank test). **There is a significant association between TTR of any type (DCIS + Invasive) and stromal Cav-1 expression ($p = 2.1 \times 10^{-4}$; log-rank test).

Table 3 Univariate association of stromal Cav-1 or inflammation with overall recurrence [ER(+) DCIS patients]

	N	Overall recurrence no N = 40	yes N = 16	p-value
Stromal Cav-1 score	56			<0.001[1]
2		78% (31)	31% (5)	
1		22% (9)	38% (6)	
0		0% (0)	31% (5)	
Stromal Cav-1 score	56			0.002[1]
2		78% (31)	31% (5)	
0/1		22% (9)	69% (11)	
Inflammation	55			0.059[1]
no		75% (30)	47% (7)	
yes		25% (10)	53% (8)	

N is the number of non-missing values. Numbers after percents are frequencies. Tests used: [1]Fisher's exact test.

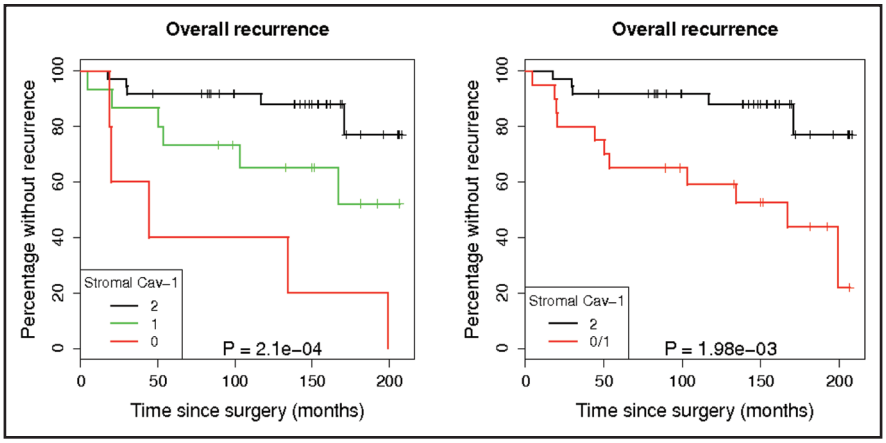

Figure 2. Kaplan-Meier curves for stromal Cav-1 status and time to overall recurrence among ER(+) DCIS patients. Note that an absence or reduction of stromal Cav-1 is associated with an increase in overall recurrence. On the left, patients were stratified into three different groups. On the right, patients with low or absent Cav-1 were considered as a single group. (Left) stromal Cav-1 (high, score = 2; low, score = 1; absent, score = 0); (Right) stromal Cav-1 (high, score = 2; low, score = 0/1). p values (log rank test) are as shown.

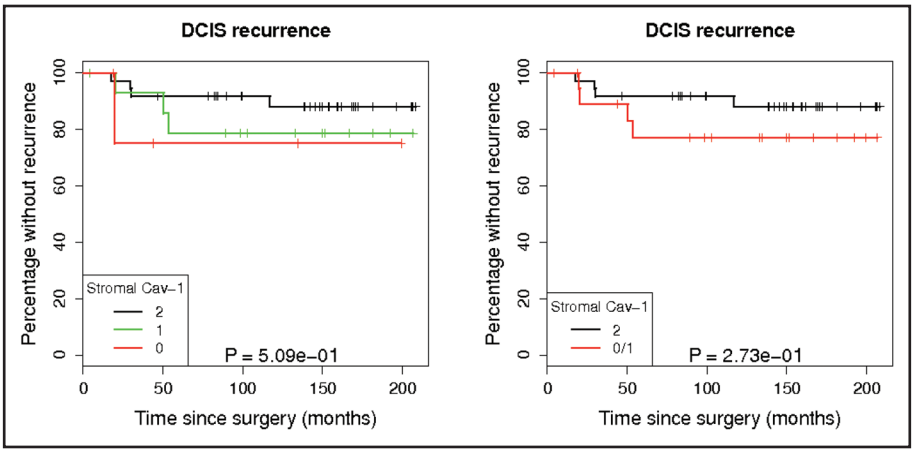

Figure 3. Kaplan-Meier curves for stromal Cav-1 status and time to overall recurrence among ER(+) DCIS patients. Note that an absence or reduction of stromal Cav-1 is not associated with an increase in DCIS recurrence. On the left, patients were stratified into three different groups. On the right, patients with low or absent Cav-1 were considered as a single group. p values (log rank test) are as shown.

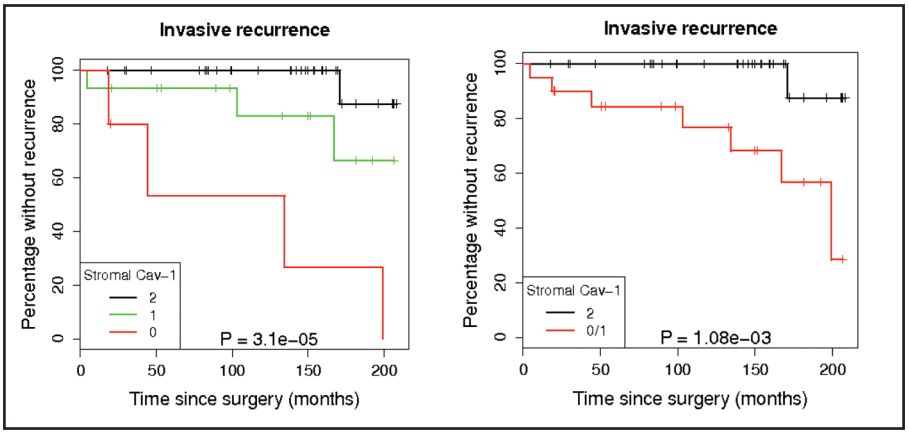

Figure 4. Kaplan-Meier curves for stromal Cav-1 status and time to overall recurrence among ER(+) DCIS patients. Note that an absence or reduction of stromal Cav-1 is strongly associated with an increase in progression to invasive breast cancer. On the left, patients were stratified into three different groups. On the right, patients with low or absent Cav-1 were considered as a single group. p values (log rank test) are as shown.

stroma of Cav-1 (-/-) null mice results in up to a 2-fold increase in tumor mass, directly demonstrating that the mammary stroma of Cav-1 (-/-) mice has tumor promoting properties.[16] As such, the mammary stromal micro-environment of Cav-1 (-/-) mice is a highly clinically-relevant animal model to genetically dissect the mechanisms underlying DCIS recurrence and invasive progression.

Genome-wide expression profiling of Cav-1 (-/-) mammary glands with DCIS-like lesions has identified the upregulation of a number of gene products normally associated with breast cancer, and/or other types of human cancer, such as *Afp, Arg1, CD276, Gipc2, Hsd17b12, Id1, Indo, Pldn, Prom1, Scg5, Spag9 and Thbs1*, among others.[15] Interestingly, many of these gene products are also known to be overexpressed in ES cells and are Nanog, Oct4, Sox2 or Myc target genes. In this group, *Prom1* (also known as CD133) is an established cell surface "cancer stem cell marker" and is often used to isolate tumor initiating cell populations. Thus, the Cav-1 (-/-) mammary stromal micro-enviroment may foster the development or propagation of epithelial cancer stem cells, leading to DCIS-like lesion formation, and local microinvasion.[15] In this regard, Cav-1 (-/-) mammary stromal fibroblasts also overexpress a subset of ES-cell related genes, suggesting that they may resemble mesenchymal stem cells;[13] similarly, human breast cancer-associated fibroblasts that lack Cav-1 protein expression, also show the overexpression of stem/progenitor cell-associated genes.[9,14] Overall, this is consistent with a model in which the mammary stromal micro-enviroment controls the "stemness" of the epithelial compartment, leading towards tumorigenesis and invasiveness.

In this paper, we show that ER(+) DCIS patients with an absence of stromal Cav-1 are at high risk for progression to invasive breast cancer. Previously, we showed that breast cancer patients with an absence of stromal Cav-1 are prone to tamoxifen-resistance.[8] By extrapolation, these ER(+) DCIS patients lacking stromal Cav-1 that progress to invasive breast cancer, would also be predicted to show tamoxifen-resistance. This may explain why the efficacy of tamoxifen-treatment in ER(+) DCIS patients is still under debate, as no significant increases in survival were previously noted for tamoxifen-treatment of ER(+) DCIS patients in at least two large multi-center clinical trials.[5,17,18] In accordance with this hypothesis, Cav-1 (-/-) null mice treated with estrogen develop mammary DCIS-like lesions which show (i) features of activated ER signaling and (ii) overexpress B23/nucleophosmin, a known marker of tamoxifen-resistance in human breast cancer patients.[15] Thus, an absence of stromal Cav-1 may explain why ER(+) DCIS "breast cancer" patients do not respond as expected to tamoxifen-treatment, as the Cav-1 negative DCIS stroma may confer a "tamoxifen-resistant" phenotype.

In summary, we show here that an absence of stromal Cav-1 in ER(+) DCIS lesions at diagnosis is highly predictive of recurrence and progression to invasive breast cancer. Conversely, DCIS patients with high stromal Cav-1 appear to be protected against invasive recurrence. As such, stromal Cav-1 may represent a valuable new biomarker for dividing DCIS patients into high-risk and low-risk groups at diagnosis, greatly facilitating their treatment stratification.

Table 4 **Univariate association of stromal Cav-1 or inflammation with DCIS recurrence (ER(+) DCIS patients)**

	N	DCIS recurrence no N = 48	DCIS recurrence yes N = 8	p-value
Stromal Cav-1 score	56			0.493[1]
2		67% (32)	50% (4)	
1		25% (12)	38% (3)	
0		8% (4)	12% (1)	
Stromal Cav-1 score	56			0.437[1]
2		67% (32)	50% (4)	
0/1		33% 16)	50% (4)	
Inflammation	55			1[1]
no		68% (32)	62% (5)	
yes		32% (15)	38% (3)	

N is the number of non-missing values. Numbers after percents are frequencies. Tests used: [1]Fisher's exact test.

Table 5 **Univariate association of stromal Cav-1 or inflammation with invasive recurrence (ER(+) DCIS patients)**

	N	Invasive recurrence no N = 48	Invasive recurrence yes N = 8	p-value
Stromal Cav-1 score	56			<0.001[1]
2		73% (35)	12% (1)	
1		25% (12)	38% (3)	
0		2% (1)	50% (4)	
Stromal Cav-1 score	56			0.002[1]
2		73% (35)	12% (1)	
0/1		27% (13)	88% (7)	
Inflammation	55			0.032[1]
no		73% (35)	29% (2)	
yes		27% (13)	71% (5)	

N is the number of non-missing values. Numbers after percents are frequencies. Tests used: [1]Fisher's exact test.

Materials and Methods

Case selection. DCIS breast tissues were obtained from the Surgical Pathology files at the Thomas Jefferson University with Institutional Review Board (IRB) approval. Seventy-eight cases for which histologic slides and blocks were available were included in the study. Clinical and treatment information was extracted by chart review. All patients were treated by the same surgeon (Gordon F. Schwartz, M.D.), undergoing surgical excision only, and all had negative margins (= or > 10 mm) at the conclusion of surgical excision or re-excision. None of the patients underwent radiation therapy, or were given tamoxifen. The date of diagnosis

Table 6 **Cox proportional hazards regression of incidence of invasive recurrence related to stromal Cav-1 expression levels**

Stromal Cav-1	N	log(HR)	Std. Err.	Hazard ratio (HR) (95% CI)	p-value
Score = 2 (ref)	36	0.00		1.00	
Score = 1	15	2.08	1.16	8.04 (0.833, 77.6)	0.07
Score = 0	5	3.52	1.13	33.86 (3.73, 307.5)	0.002
Stromal Cav-1	**N**	**log(HR)**	**Std. Err**	**Hazard ratio (HR) (95% CI)**	**p-value**
Score = 2 (ref)	36	0.0		1.0	
Score = 0/1	20	2.66	1.07	14.24 (1.75, 116.0)	0.01

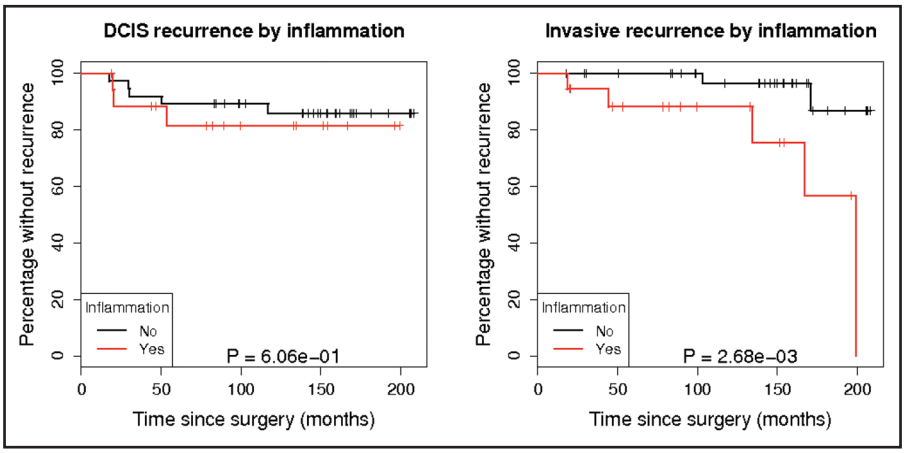

Figure 5. Kaplan-Meier curves for inflammation and time to DCIS recurrence and invasive recurrence among ER(+) DCIS patients. Note that inflammation is associated with an increase in progression to invasive breast cancer, but not simple DCIS recurrence. p values (log rank test) are as shown.

and the date of recurrence were defined as the date of surgery that established the pathologic diagnosis. Local recurrence of DCIS or invasive breast carcinoma or absence of disease at the last follow-up constituted an endpoint with respect to data analysis. Hematoxylin and eosin-stained slides were reviewed for all cases and pathological variables were determined following well-established criteria.[10,11] For each case, histologic pattern (solid, cribriform, papillary, micropapillary, comedo), nuclear grade, presence of necrosis and inflammation were recorded. Results of immunohistochemical stains for estrogen receptor (ER), progesterone receptor (PR) and HER2 were available for a subset of cases.

Patients. An initial cohort of 78 DCIS patients with follow-up data was examined. None of the patients with ER-negative DCIS suffered recurrence; all of the recurrences occurred in the patients who were ER-positive. We, therefore, focused our analysis on this subset of 56 patients. There were 55 women and one male diagnosed with DCIS, with a median age of 55 years (range 32–87 years). The median follow-up time for these patients was 146.8 months (12.23 years). Sixteen had a recurrence during follow-up with eight cases recurring as DCIS and eight as invasive breast carcinoma. The median time to recurrence was 47.3 months (29.88 months for DCIS, 118.7 months for invasive breast cancer).

Immunohistochemistry. Cav-1 expression in the tumor stroma was assessed by employing a standard immuno-peroxidase method. Sections were de-paraffinized and went through antigen retrieval using high pH retrieval buffer (Dako, Carpinteria, CA) before being stained using a Dako autostainer. Endogenous peroxidase activity was blocked using Dako FLEX Peroxidase block; sections were incubated for 30 min with the primary antibody (mouse monoclonal antibody (clone #2297); anti-caveolin-1 IgG, BD Biosciences, San Jose, CA; 1:50 dilution) followed by washing three times with TBS. Sections were incubated for 20 min with HRP polymer (Dako) and subsequently washed three times before applying the DAB chromogen with substrate. Sections were washed three times before counter-stained with Hematoxylin. Finally, sections went through dehydration

Table 7 **Association of stromal Cav-1 and recurrence, in the presence of inflammation**

	Stromal Cav-1 status			
	Score = 2 N = 6	Score = 1 N = 7	Score = 0 N = 5	p-value
Rec. Type				0.003
None	100% (6)	57% (4)	0% (0)	
DCIS	0% (0)	29% (2)	20% (1)	
Invasive	0% (0)	14% (1)	80% (4)	

Only patients with inflammation were selected for this analysis. Note that, in the presence of inflammation, high stromal Cav-1 levels are still predictive of no recurrence, while absent stromal Cav-1 is predictive of invasive recurrence. Numbers after percents are frequencies. Test used: Fisher's exact test.

Table 8 **Association of stromal Cav-1 and recurrence, in the absence of inflammation**

	Stromal Cav-1 status		
	Score = 2 N = 30	Score = 1 N = 7	p-value
Rec. Type			0.514
None	83% (25)	71% (5)	
DCIS	13% (4)	14% (1)	
Invasive	3% (1)	14% (1)	

Only patients without inflammation were selected for this analysis. Numbers after percents are frequencies. Test used: Fisher's exact test

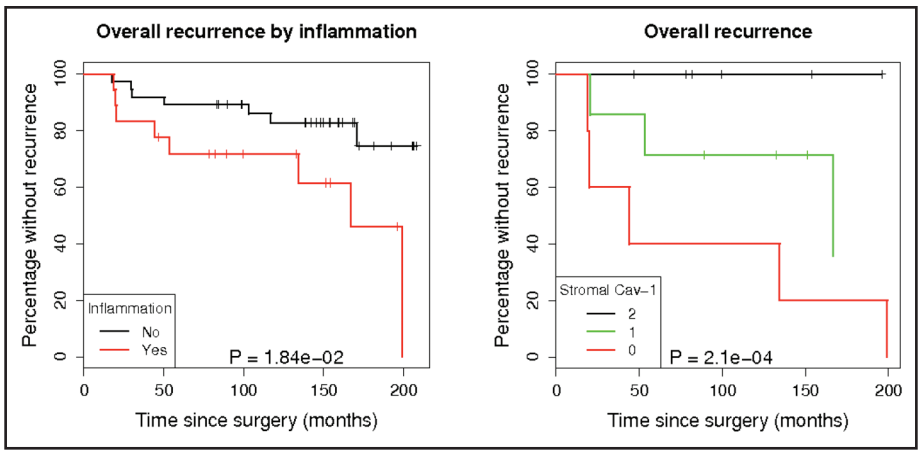

Figure 6. Kaplan-Meier curves for Inflammation, stromal Cav-1 status, and time to overall recurrence among ER(+) DCIS patients. (Left) Note that inflammation is associated with an increase in overall recurrence. (Right) However, this effect is still strictly dependent on Cav-1 levels. For example, high stromal Cav-1 levels are protective against overall recurrence, even in the presence of inflammation (see also Tables 7 and 8).

before being mounted. The staining was scored semi-quantitatively as negative (0; no staining), weak (1; either diffuse weak staining or strong staining in less than 30% of stromal cells) and strong (2; defined as strong staining of 30% or more of the stromal cells).

Statistics. Each patient has data on age, date of surgery for primary DCIS, presence of recurrence (DCIS or breast cancer) and the date recurrence was detected, the DCIS type, nuclear grade, presence of necrosis and presence of inflammation. ER and PR status was available for 87% (68/78) of the patients, and HER2 status was available for 32% (25/78) of the patients. Stromal Cav-1 was scored for each tissue sample using a numeric score of 0, 1 or 2, depending on the degree of stromal Cav-1 staining. A score of 0 was interpreted as absence of stromal Cav-1, and scores of 1 and 2 were interpreted as low and high levels of stromal Cav-1, respectively.

Our primary outcome in this study is recurrence of DCIS or progression to invasive ductal carcinoma, after diagnosis with primary DCIS. Since no patient in the study had a recurrence of both DCIS and invasive carcinoma, we considered the two recurrence processes separately for analysis, accounting for competing risks. A patient who had DCIS recurrence was considered censored at the time of recurrence for the invasive carcinoma recurrence analysis, and vice versa. The time to recurrence (TTR) from time of surgery was evaluated using Kaplan-Meier survival curves and comparison of stratified survival curves using log-rank tests. The cumulative incidence of recurrence of each type (DCIS and invasive carcinoma) was computed taking into account competing risks.[12] We computed 15-year recurrence rates based on the Kaplan-Meier estimates of the TTR. Cox proportional hazards regression was used to estimate the association between stromal Cav-1 levels and TTR, both univariately and multivariately, in the presence of other potential prognostic factors for TTR. We also investigated the association between stromal Cav-1 levels and other prognostic factors to identify potential confounders. All hypothesis testing was performed using the Fisher exact test or the Kruskal-Wallis test, depending on the discrete or continuous nature of the other factors.

All p-values were two-sided, and $p < 0.05$ was considered statistically significant. Statistical analysis was performed, and tables and graphs generated, using the R statistical analysis software version 2.8.1.

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Note

Supplementary materials can be found at:
www.landesbioscience.com/supplement/
WitkiewiczCBT8-11-Sup.pdf

References

1. Jemal A, Siegel R, Ward E, Hao Y, Xu J, Murray T, et al. Cancer statistics, 2008. CA Cancer J Clin 2008; 58:71-96.
2. Jemal A, Siegel R, Ward E, Murray T, Xu J, Thun MJ. Cancer statistics, 2007. CA Cancer J Clin 2007; 57:43-66.
3. Silverstein MJ, Lagios MD, Martino S, Lewinsky BS, Craig PH, Beron PJ, et al. Outcome after invasive local recurrence in patients with ductal carcinoma in situ of the breast. J Clin Oncol 1998; 16:1367-73.

4. Allred DC, Bryant J, Land S, Paik S, Fisher E, Julian T, et al. Estrogen receptor expression as a predictive marker of the effectiveness of tamoxifen in the treatment of DCIS: findings from NSABP Protocol B-24. Breast Cancer Res Treat 2002; 76:36.
5. Houghton J, George WD, Cuzick J, Duggan C, Fentiman IS, Spittle M. Radiotherapy and tamoxifen in women with completely excised ductal carcinoma in situ of the breast in the UK, Australia and New Zealand: randomised controlled trial. Lancet 2003; 362:95-102.
6. Sanders ME, Schuyler PA, Dupont WD, Page DL. The natural history of low-grade ductal carcinoma in situ of the breast in women treated by biopsy only revealed over 30 years of long-term follow-up. Cancer 2005; 103:2481-4.
7. Consensus Conference on the classification of ductal carcinoma in situ. The Consensus Conference Committee. Cancer 1997; 80:1798-802.
8. Witkiewicz AK, Dasgupta A, Sotgia F, Mercier I, Pestell RG, Sabel M, et al. An absence of stromal caveolin-1 expression predicts early tumor recurrence and poor clinical outcome in human breast cancers. Am J Pathol 2009; 174:2023-34.
9. Witkiewicz AK, Casimiro MC, Dasgupta A, Mercier I, Wang C, Bonuccelli G, et al. Towards a new "stromal-based" classification system for human breast cancer prognosis and therapy (extra-views). Cell Cycle 2009; 8:1654-8.
10. Balleine RL, Webster LR, Davis S, Salisbury EL, Palazzo JP, Schwartz GF, et al. Molecular grading of ductal carcinoma in situ of the breast. Clin Cancer Res 2008; 14:8244-52.
11. Cornfield DB, Palazzo JP, Schwartz GF, Goonewardene SA, Kovatich AJ, Chervoneva I, et al. The prognostic significance of multiple morphologic features and biologic markers in ductal carcinoma in situ of the breast: a study of a large cohort of patients treated with surgery alone. Cancer 2004; 100:2317-27.
12. Kalbfleisch JD, Prentice RL. The statistical analysis of failure time data. New York: John Wiley & Sons 2002.
13. Sotgia F, Del Galdo F, Casimiro MC, Bonuccelli G, Mercier I, Whitaker-Menezes D, et al. Caveolin-1$^{-/-}$ null mammary stromal fibroblasts share characteristics with human breast cancer-associated fibroblasts. Am J Pathol 2009; 174:746-61.
14. Mercier I, Casimiro MC, Wang C, Rosenberg AL, Quong J, Allen KG, et al. Human breast cancer-associated fibroblasts (CAFs) show caveolin-1 downregulation and RB tumor suppressor functional inactivation: implications for the response to hormonal therapy. Cancer Biol Ther 2008; 7:1212-25.
15. Mercier I, Casimiro MC, Zhou J, Wang C, Plymire C, Bryant KG, et al. Genetic ablation of caveolin-1 drives estrogen-hypersensitivity and the development of DCIS-like mammary lesions. Am J of Pathol 2009; 174:1172-90.
16. Williams TM, Sotgia F, Lee H, Hassan G, Di Vizio D, Bonuccelli G, et al. Stromal and epithelial caveolin-1 both confer a protective effect against mammary hyperplasia and tumorigenesis: caveolin-1 antagonizes cyclin D1 function in mammary epithelial cells. Am J Pathol 2006; 169:1784-801.
17. Leonard GD, Swain SM. Ductal carcinoma in situ, complexities and challenges. J Natl Cancer Inst 2004; 96:906-20.
18. Fisher B, Dignam J, Wolmark N, Wickerham DL, Fisher ER, Mamounas E, et al. Tamoxifen in treatment of intraductal breast cancer: National Surgical Adjuvant Breast and Bowel Project B-24 randomized controlled trial. Lancet 1999; 353:1993-2000.